Science and economics · Question booklet

Physics D.3 · Chemistry S2.3 · Economics 3.6

1. Measuring the curvature of an ion beam

Physics SL / HL · D.3 · Paper 1B-style data analysis · 11 marks

A narrow beam of positive ions enters a uniform magnetic field at right angles. Its speed is fixed at 4.00×10^{5} m s^{-1}. A camera records the radius of the circular arc. Assume the field and speed uncertainties are negligible. Each radius has an uncertainty of ±0.2 cm. The readings are simulated.

Magnetic field / mT	20.0	25.0	40.0	50.0
Radius / cm	10.0	8.1	5.0	4.1

a. Use the 25.0 mT reading to calculate the magnitude of the charge-to-mass ratio of an ion.
[3]

b. Use the first and last readings and their uncertainties to assess the prediction that radius is inversely proportional to field strength.
[3]

c. The beam initially travels to the right and bends upwards on the camera image. Determine the direction of the field relative to the image. Explain why the speed is constant along the arc.
[3]

d. At larger fields the arcs are smaller. Suggest one change to the imaging procedure that would reduce the fractional uncertainty in their radii.
[2]

2. Finding a camera offset by reversing a field

Physics SL / HL · D.3 · Paper 1B-style data analysis · 12 marks

A positive-ion beam travels horizontally at 2.00×10^{5} m s^{-1} between parallel plates of length 0.0800 m and separation 0.0200 m. Ignore gravity and fringing. With the plate voltage reversed the electric field reverses. The signed camera reading at the plate exit includes an unknown constant offset. Positive displacement is upwards. Each reading has uncertainty ±0.10 mm. The data are simulated.

Potential difference / V	−40	−20	+20	+40
Camera reading / mm	−4.30	−1.90	+2.90	+5.30

a. Use the two 40 V readings to determine the camera offset and the magnitude of the beam displacement caused by a 40 V potential difference.
[3]

b. Determine the charge-to-mass ratio of the ions using the corrected displacement.
[4]

c. Use the two 20 V readings to test a prediction of the uniform-field model.
[2]

d. Estimate the maximum absolute uncertainty in the corrected 4.80 mm displacement from the two readings. Explain whether taking many pairs would eliminate an error in the stated plate separation.
[3]

3. A crossed-field selector with a residual field

Physics SL / HL · D.3 · Paper 1B-style data analysis · 9 marks

Positive ions pass through perpendicular electric and magnetic fields at right angles to their velocity. For each applied coil setting a student adjusts an opposing electric field until the beam shows no deflection. The applied magnetic field is accurate. A small, constant background magnetic field may also act along the coil axis. Electric-field readings have uncertainty ±0.10 kV m^{-1}. The readings are simulated.

Applied magnetic field / mT	4.0	6.0	8.0	10.0
Balancing electric field / kV m^{-1}	2.40	3.40	4.40	5.40

a. Using the end readings, determine the gradient of balancing electric field against applied magnetic field in SI units and interpret it physically.

[3]

b. Estimate the background field, including its direction relative to the applied field.

[3]

c. A second student says the non-zero intercept proves that there is a background magnetic field. Evaluate this claim and suggest a measurement that would distinguish a background field from an offset in the electric-field reading.

[3]

4. Melting points within an alkali-metal group

Chemistry SL / HL · S2.3 · Paper 1B-style data analysis · 8 marks

The table gives rounded melting points of pure elements at ordinary pressure. The metallic model represents each of these Group 1 metals by singly charged positive ions and delocalized electrons. The ion radius increases from lithium to potassium. Reference values are rounded from the Royal Society of Chemistry periodic table.

Metal	Lithium	Sodium	Potassium
Melting point / K	453.7	370.9	336.5

a. Calculate the percentage decrease in melting point from lithium to potassium using the kelvin scale.
[2]

b. Account for the trend using the metallic model.
[3]

c. A student says that a molten sample of potassium cannot conduct because its lattice has collapsed. Evaluate this prediction.
[3]

5. Which observations identify a metallic solid?

Chemistry SL / HL · S2.3 · Paper 1B-style data analysis · 10 marks

Three pure, sealed samples of matched dimensions were tested at 25 °C and after melting. Each current was measured with the same applied potential and electrode separation. These simulated tests were conducted under inert conditions. An entry below 0.01 mA means the reading is below the detection limit. Only sample P could be rolled into a thin sheet without fracturing.

Sample	Current, solid / mA	Current, liquid / mA	Melting point / °C
P	36.0	20.0	232
Q	<0.01	7.20	801
R	<0.01	<0.01	81

a. Use the measurements to calculate the ratio of the solid current to the liquid current for P. Deduce whether P is still a conductor after melting.

[2]

b. Use the metallic model to explain both the conductivity of P as a solid and its response to rolling.

[3]

c. A student identifies Q as a metal because its melting point exceeds that of P. Using two observations, assess this conclusion and suggest a more plausible structure for Q.

[3]

d. Explain why the readings alone do not establish that R has absolutely zero conductivity.

[2]

6. Resolving two close melting temperatures

Chemistry SL / HL · S2.3 · Paper 1B-style data analysis · 10 marks

A school uses one calibrated sensor and the same heating procedure for pure samples of magnesium and aluminium. Each row contains three independent estimates of the onset of melting, in °C. The sensor can have a common offset of up to ±3 °C that is unchanged between metals. These readings are simulated.

Metal	Run 1	Run 2	Run 3
Magnesium	648	651	651
Aluminium	658	661	661

In the simple metallic model, magnesium supplies two outer electrons per atom and aluminium supplies three. The aluminium ion core is smaller.

a. Calculate the mean melting temperature and the half-range of the three readings for each metal.
[3]

b. Use the results to judge whether the common sensor offset could explain the observed difference. Distinguish repeatability from accuracy.
[3]

c. Explain the direction of the difference using the metallic model. Assess a claim that the third outer electron alone accounts for the measured size of the difference.
[4]

7. Can household transfers close a recessionary gap?

Economics SL / HL · 3.6 · Paper 1 essay option · 25 marks

Answer both parts. Use a labelled economic diagram where appropriate and draw on a real economy you have studied.

a. Explain how a temporary transfer payment to households can help close a recessionary output gap.
[10]

b. Using real-world examples, evaluate the effectiveness of expansionary fiscal policy in raising real output during a recession.
[15]